

Action–Angle Components, Exact Return Orders, and Zeta Laws for Every Periodic Box–Ball Level Set

Route-A structural certificate C182

Abstract

For the periodic binary box–ball system, we freeze every $L \geq 2M$, every soliton content and internal-symmetry sector, and every commuting carrier evolution T_l . The action–angle bijection and invariant-torus decomposition are prior theorems of Kuniba–Takagi–Takenouchi and Takagi, not novelty claims. From their precise component matrix we derive a single all-parameter theorem: augmented Smith normal form gives the exact order of each T_l translation; component formulas aggregate over symmetries, contents, fixed mass, and the full positive-weight space; Möbius inversion gives primitive cycles; and the Artin–Mazur zeta is the reciprocal ordinary determinant of the finite counting-measure Koopman permutation. Vacuum, half filling, and saturated carrier capacity are included. An independent periodic-carrier enumeration checks every binary state through $L = 9$. The exact cycles have no intrinsic rational-prime origin or logarithmic arithmetic clock, so the candidate is mathematically retained and Route-A rejected.

Keywords: periodic box–ball system; action–angle variables; internal symmetry; Smith normal form; dynamical zeta; Koopman permutation.

中文摘要

本文固定周期二元盒球系统的全部参数： $L \geq 2M$ 、任意孤子含量、任意内部对称扇区以及全部可交换载流子演化 T_l 。作用角变量双射与不变环面分解属于 Kuniba–Takagi–Takenouchi 和 Takagi 的既有定理，本文不声称其新颖性。在该定理上，我们用增广 Smith 标准形给出每个平移的精确阶，把不动点公式逐层汇总到全部对称、全部孤子含量、固定球数和完整正权状态空间；再由莫比乌斯反演得到本原周期，并证明动力 ζ 函数等于有限计数测度 Koopman 置换普通行列式的倒数。真空、半填充和载流子饱和和边界均被保留。独立的周期载流子程序穷举核验了 $L \leq 9$ 的所有二元状态。由于这些精确周期没有内生素数来源或对数算术时钟，数学结果被保留，但 Route-A 候选被否决。

关键词：周期盒球系统；作用角变量；内部对称；Smith 标准形；动力 ζ 函数；Koopman 置换。

1 Frozen source dynamics and action data

Let $\mathcal{P}_{L,M}$ be the binary words on a ring of length L with M balls, where $0 \leq M \leq \lfloor L/2 \rfloor$. The capacity- l periodic combinatorial- R carrier defines T_l ; one application is one unit of the discrete physical clock. KTT prove that the T_l commute and preserve the soliton content [1]

$$m = (m_j)_{j \geq 1}, \quad \sum_{j \geq 1} j m_j = M.$$

Put $H = \{j : m_j > 0\}$ and

$$p_j = L - 2 \sum_{k \in H} \min(j, k) m_k. \quad (1)$$

Since $L \geq 2M$, every $p_j \geq 0$; for $N = \max H$, $p_N = L - 2M$. Thus equality in the density bound is visible as $p_N = 0$, rather than excluded.

For every $j \in H$, choose

$$\alpha_j \mid \gcd(m_j, p_j), \quad \gcd(m, 0) = m,$$

and order H increasingly. Define

$$F_{jk}^{(\alpha)} = \frac{\delta_{jk} p_k + 2 \min(j, k) m_k}{\alpha_k}, \quad h_l = (\min(j, l))_{j \in H}. \quad (2)$$

The KTT/Takagi component theorem states that every connected component of the symmetry- α level is equivariantly bijective to [2]

$$G_\alpha = \mathbb{Z}^H / F^{(\alpha)} \mathbb{Z}^H, \quad T_l[x] = [x + h_l]. \quad (3)$$

It has $D_\alpha = \det F^{(\alpha)} > 0$ points.

The exact number of copies of (3) is

$$c(\alpha) = \prod_{j \in H} \frac{|\Lambda^{(\alpha_j)}(m_j, p_j)|}{m_j / \alpha_j}, \quad (4)$$

where

$$|\Lambda^{(\alpha)}(m, p)| = \sum_{\substack{\beta | \gcd(m, p) \\ \alpha | \beta}} \mu(\beta / \alpha) \binom{(p + m) / \beta - 1}{m / \beta - 1}. \quad (5)$$

Equation (5) is Möbius inversion by exact internal period. The denominator in (4) is integral because cyclic shift acts freely with orbit size m / α on an exact-period sequence. The sector identity is

$$\Omega_L(m) = \sum_{\alpha} c(\alpha) D_{\alpha} = \det F^{(1)} \prod_{j \in H} \frac{1}{m_j} \binom{p_j + m_j - 1}{m_j - 1}. \quad (6)$$

These statements are imported with attribution. Our contribution starts from the exact finite translations in (3) and derives their complete fixed-cycle-determinant consequences.

2 Augmented Smith order

Fix one component, abbreviate $F = F^{(\alpha)}$, $h = h_l$, and $D = \det F$, and let $B = [F \mid h]$. Write the nonzero Smith factors of B as $e_1, \dots, e_s$.

Theorem 1 (Exact component return order). *The order of T_l on the component is*

$$q_{\alpha, l} = \min\{n \geq 1 : nh_l \in F^{(\alpha)} \mathbb{Z}^H\} = \frac{D_{\alpha}}{e_1 \cdots e_s}. \quad (7)$$

If $UFV = \text{diag}(d_1, \dots, d_s)$, equivalently

$$q_{\alpha, l} = \text{lcm}_i \frac{d_i}{\gcd(d_i, (Uh_l)_i)}.$$

Indeed, quotienting G_{α} by the cyclic subgroup generated by $[h]$ gives

$$G_{\alpha} / \langle [h] \rangle \cong \mathbb{Z}^s / (F\mathbb{Z}^s + \mathbb{Z}h).$$

The group on the right has size $e_1 \cdots e_s$ by Smith normal form, while $|G_{\alpha}| = D$. Their ratio is the order of the cyclic subgroup, proving (7). The second formula is the coordinate divisibility condition after applying U . No determinant ratio is used without its lattice-index justification.

Since $h_l = (j)_{j \in H}$ once $l \geq \max H$, all larger carriers induce the same translation on this level. If $H = \emptyset$, the empty determinant and empty Smith product are one; the vacuum component is a singleton with $q = 1$.

3 Fixed points, aggregation, and primitive cycles

Translation by an element of order q on a finite group has a fixed point at iterate n precisely when $q \mid n$, in which case every point is fixed. Therefore one component satisfies

$$\# \text{Fix}(T_l^n | G_{\alpha}) = D_{\alpha} \mathbf{1}_{q_{\alpha, l} | n}. \quad (8)$$

After multiplying by the number of copies, the entire soliton level has

$$F_{m, l}(n) = \sum_{\alpha} c(\alpha) D_{\alpha} \mathbf{1}_{q_{\alpha, l} | n}. \quad (9)$$

Because the action variable partitions the binary words,

$$F_{L, M, l}(n) = \sum_{m: \sum j m_j = M} F_{m, l}(n), \quad F_{L, l}^+(n) = \sum_{M=0}^{\lfloor L/2 \rfloor} F_{L, M, l}(n), \quad (10)$$

with state totals $\binom{L}{M}$ and $\sum_{M \leq L/2} \binom{L}{M}$, respectively. Thus “all-state” means an exact finite aggregation, not a sampled orbit list.

Let P_n be the number of points of exact period n and $C_n = P_n / n$ the number of primitive cycles. Möbius inversion gives

$$P_n = \sum_{d | n} \mu(n/d) F(d), \quad C_n = P_n / n. \quad (11)$$

Equivalently, one component contributes $D_{\alpha} / q_{\alpha, l}$ cycles of length $q_{\alpha, l}$, multiplied by $c(\alpha)$. This proves integrality and completeness without a periodic-orbit cutoff.

4 Zeta and the finite Koopman determinant

For any one of the aggregations above, let C_q denote its number of length- q cycles. Equations (8)–(11) give

$$\zeta_{T_l}(z) = \exp\left(\sum_{n \geq 1} \frac{F(n)}{n} z^n\right) = \prod_q (1 - z^q)^{-C_q}. \quad (12)$$

On $\ell^2(X)$ with counting measure, $U_l f = f \circ T_l$ is a canonical finite unitary permutation. A q -cycle has characteristic polynomial $x^q - 1$, hence

$$\det(I - zU_l) = \prod_q (1 - z^q)^{C_q} = \zeta_{T_l}(z)^{-1}. \quad (13)$$

This is the ordinary determinant of the same-clock finite operator. It is neither a regularized infinite-volume determinant nor a Hilbert–Pólya construction.

5 Boundary closure

At half filling $p_N = 0$. In (5), all binomials for that coordinate equal one, so Möbius cancellation leaves only $\alpha_N = m_N$ with value one. Thus the component theorem remains finite at $L = 2M$. At $M = 0$, the one-point vacuum is fixed by every T_l . At $l \geq \max H$, the vector h_l saturates. These calculations close the three singular-looking edges without changing the clock or deleting states.

6 Independent certificate and integrity audit

The exact regression ledger exhausts $2 \leq L \leq 14$: 193 soliton levels, 236 positive internal-symmetry sectors, 1,524 translations, 342 fixed-mass aggregates, 62 positive-weight aggregates, and 23,136 fixed-point cells. For every level, the sector sum $\sum_\alpha c(\alpha) D_\alpha$ equals (6); summing all contents equals $\binom{L}{M}$.

The checker imports no producer code. Rational elimination and independent determinantal divisors pass 55,907 assertions. More importantly, it runs the actual periodic carrier on every one of 559 states through $L = 9$, checks 108 complete permutations, reconstructs soliton content from conserved energies, and matches every level and all-state cycle spectrum. SymPy passes 38,979 determinant, Smith, order-minimality, Möbius, primitive-cycle, and cycle-determinant checks. Producer replay is byte exact; 64 repaired-hash semantic mutations and one stale-hash mutation are rejected.

The seven mandatory integrity modes are all **CLEAR**: (1) implementation bugs are attacked by independent algebra and carrier enumeration; (2) both citations and formula ownership were source-verified; (3) every reported number has a deterministic command; (4) no data shortcut replaces the all-parameter proof; (5) no unexpected output is reframed as novelty; (6) methods match the executed scripts; and (7) frame-lock was tested by the precommitted Brockett fallback, which was unnecessary only after the full internal-symmetry theorem closed. This is an internal integrity audit, not external peer review.

7 Route-A decision

Primitive cycles are intrinsic and complete but contain no arithmetic information or stability/von-Mangoldt weights: **A1_WEAK**. The source zeta and finite determinant are exact but make no target-divisor comparison: **A2_FAIL**. There is no target functional equation, counting law, continuation, or Weil compression: **A3_FAIL**. The same-clock counting-measure permutation is a natural finite quantization: **A4_NATURAL_QUANTIZATION**. None repairs the absence of an intrinsic rational-prime origin or arithmetic clock: **A0_FAIL**. Hence

$$(\text{A0_FAIL}, \text{A1_WEAK}, \text{A2_FAIL}, \\ \text{A3_FAIL}, \text{A4_NATURAL_QUANTIZATION}),$$

overall **ROUTE_A_REJECTED**; Route B is false.

Limitations. The imported action–angle result is not rebranded as a new theorem. No canonical infinite-volume normalization, arithmetic primitive weight, target divisor, Gamma factor, functional equation, Riemann–von Mangoldt law, Weil compression, arithmetic local factor, Euler factor, root number, automorphy claim, or Hilbert–Pólya operator is supplied. Scope: **NO_BAD_EULER_OR_ROOT_NUMBER**.

References

- [1] A. Kuniba, T. Takagi, and A. Takenouchi, “Bethe ansatz and inverse scattering transform in a periodic box-ball system,” *Nuclear Physics B* 747 (2006), 354–397, arXiv:math/0602481v2.
- [2] T. Takagi, “Level Set Structure of an Integrable Cellular Automaton,” *SIGMA* 6 (2010), 027, arXiv:0906.1410.

Declarations

Data and code availability. Package-local exact evidence and deterministic code accompany this manuscript.

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